

## **Integrating Multinomial Logit and Machine Learning Algorithms to Detect Crop Choice Decision Making**

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**Abstract**— In this study, we explore the influence of environmental and soil factors on crop suitability using statistical models and machine learning techniques. We employ a Multinomial Logit Model (MLM) and predictive models including Random Forest (RF), Gradient Boosting (GB), and Light Gradient Boosting Machine (LightGBM) to assess their impact on rice and maize production. Our findings indicate that nitrogen, rainfall, and humidity significantly enhance rice yields, whereas temperature and soil pH negatively affect it. For maize, nitrogen is beneficial, while potassium, temperature, rainfall, and soil pH are detrimental. The models also highlight the paramount importance of rainfall and humidity in crop selection, with both factors having substantial importance scores across RF, GB, and LightGBM. This data-driven approach achieves over 99% accuracy in crop classification via RF, suggesting a robust framework for agricultural decision-making that can significantly improve crop productivity and sustainability.

**Index Terms**—Gradient boosting, LightGBM, multinomial logit, crop choice decision-making, agro environmental factors

### **I. INTRODUCTION**

Agriculture has long been the backbone of human civilization, with crop selection being a pivotal decision that directly impacts both food security and economic stability [1, 3, 7]. In the modern era, this age-old decision-making process has been transformed by the advent of data-driven methodologies, presenting an opportunity to align crop choices with environmental sustainability and optimal resource utilization.

The complexity of factors influencing crop viability—from soil chemistry to climate patterns—necessitates a multi-faceted approach to agricultural planning [8, 10, 11]. Existing studies mentioned agricultural farmers' adoption decision behavior by analyzing primary survey data including factors relevant to producer perception, adoption decision, and economic and socio-

demographic issues [4, 19]. It is useful to evaluate the economic value and economic risk associated with farm activities by analyzing different management decision criteria

[9]. A study presented the effects of farmer demographics (i.e., age, education, gender, farm income, non-farm income, and marital status), farm size, storage capacity, and access to a rural agricultural extension on agricultural farmers' crop marketing decisions [5]. Also, the effects on crop choice decisions due to changes in crop prices, oil and biofuel volume and prices, and other exogenous variables (e.g., climate change, market demand for ethanol) are evident as per a few studies conducted on North Dakota farmers [12, 13, 18, 26]. Further, a few studies also indicated that weather variables (e.g., precipitation, average temperature) along with lagged yields of field crops, and crop prices have an impact on field crop yield [2, 6, 23, 27, 28]. Additionally, existing literature based on South Dakota producers indicated high satisfaction with adopting conservation measures to their farmland and conservation attitudes on agro environmental conditions [16, 22, 25].

Traditional methods rely on historical knowledge and trial and error, but these can fall short in the face of changing climatic conditions and evolving agricultural practices. The challenge lies in systematically identifying and quantifying the myriad factors that dictate the suitability of one crop over another in a given locale [29–34].

This research addresses the critical need for precision agriculture by harnessing the power of statistical analysis and machine learning. We propose a novel dual-approach methodology that integrates rigorous statistical testing with predictive modeling, aiming to discern the key environmental and soil factors that most significantly influence crop choice decision-making. By leveraging statistical analysis of variance (ANOVA), we examine the variance in crop yield data across different conditions, identifying factors with significant impacts. As per the multinomial logit model, we also find that agro environmental factors play a critical role in crop choice decision-making. The MLM is instrumental in our re- search for understanding the specific impacts of various agro environmental factors on crop decision-making. By analyzing how factors like nitrogen, rainfall, and humidity affect the likelihood of choosing rice over maize, we quantify their influences. This model allows us to directly interpret the significance and effect size of each variable, providing clear, actionable insights for agricultural stakeholders. Concurrently, we deploy machine learning algorithms, renowned for their predictive prowess, to model and predict crop selection based on these factors. The integration of these methodologies enables us to not only confirm the relevance of identified factors but also to predict their combined effect on crop suitability with a high degree of accuracy through.

The significance of this study is twofold. Firstly, it offers a data-centric foundation for agricultural decision support systems, thereby enhancing the precision and reliability of crop selection strategies. Secondly, it contributes to the body of knowledge by providing a comprehensive analysis of agro environmental factors through the lens of advanced statistical and computational

techniques. As the agricultural sector stands on the brink of a technological revolution, our research is poised to facilitate a paradigm shift from traditional intuition-based methods to analytics-driven crop selection, ushering in an era of increased productivity and environmental harmony. In the following sections, we detail our methodological framework, present our key findings, and discuss their practical implications for the agricultural community, ultimately contributing a model of analysis that is both replicable and scalable across diverse agricultural landscapes.

## **II. LITERATURE REVIEW**

Based on recent literature in the realm of agricultural technology, there's an increasing focus on integrating machine learning (ML) with crop selection systems [14, 15, 17, 20, 21]. Notably, current advancements have made strides in combining weather and soil parameters to recommend optimal crops for specific land areas. This has been facilitated by the use of advanced ML techniques such as Recurrent Neural Networks for weather prediction and Random Forest Classification for crop selection, taking into account both current and historical data.

The improvements we are bringing in this paper include a holistic approach that not only considers a wide array of environmental and soil factors but also employs a dual analytical methodology that integrates both multinomial logit marginal effects analysis and advanced machine learning models for enhanced prediction accuracy and interpretability. Our research bridges the gap by providing a comprehensive analysis that can inform more data-driven and precise cultivation strategies. For example, [24] paper finds the accuracy of crop prediction 80% while with an extensive dataset the prediction of crop is 99% in our case.

This work represents an evolution in crop selection strategies, promising to contribute significantly to the field by offering a nuanced understanding of the interplay between multiple critical factors and their collective impact on agricultural decision-making.

## **III. METHODOLOGY**

### **A. Data Acquisition and Preprocessing**

The foundation of this study rests on an extensive dataset encompassing agro environmental factors pertinent to crop choice decision-making. The dataset was curated from Kaggle [25], ensuring a comprehensive compilation of variables such as crop choices, nitrogen (N), phosphorus (P), potassium (K) content, temperature, humidity, soil pH, and rainfall. Rigorous preprocessing was undertaken to ensure data integrity, including normalization and handling of missing values through advanced imputation techniques. The dataset has 2200 entries. The data has been splitted to training and test dataset for 80% and 20% respectively.

### **B. Statistical Analysis- Multinomial Logistic Regression**

we employ a multinomial logit model to estimate the effects of response variables (e.g. Nitrogen, Phosphorus, Potassium, temperature, humidity, soil pH, rainfall) on predictor variables (crop choice). We also reported an Analysis of Variance (ANOVA) to statistically evaluate the impact of each factor on crop yield variability. This analysis enabled us to discern the significant contributors among the considered factors, setting the stage for further predictive modeling.

### C. Machine Learning Model Development

Leveraging the insights from the statistical analysis, we constructed several machine-learning models. For this study we considered three ML models for comparison so that the impact can be understood better. The Models are:

- 1) Random Forest (RF): An ensemble learning method known for its robustness, handling both classification and regression tasks effectively. RF was utilized to assess feature importance and model the non-linear relationships between variables.
- 2) Gradient Boosting (GB): A powerful boosting algorithm that builds on weak learners sequentially to minimize errors. Its utilization was aimed at improving predictive performance incrementally.
- 3) Light Gradient Boosting Machine (LGBM): A gradient boosting framework that employs tree-based learning algorithms optimized for speed and efficiency, ideal for large datasets.

Each model was trained and validated using a split of training and testing data to ensure the model's ability to generalize beyond the observed data. Hyperparameter tuning was performed via grid search and cross-validation techniques to optimize the models' performance.

### D. Feature Importance Analysis

Post-training, a feature importance analysis was conducted for each model to identify and rank the most influential factors in crop type classification. This analysis not only corroborated the statistical findings but also provided a granular view of the relative importance of each factor, facilitating a deeper understanding of their individual and collective influence on crop selection.

### E. Model Evaluation

Model performance was assessed using a suite of metrics including accuracy, precision, recall, and the F1 score. These metrics provided a holistic view of the model's predictive capabilities. The evaluation was designed to ensure robustness and reliability, with a particular focus on the models' applicability in practical agricultural scenarios.

The ML models used in our study, such as Random Forest, Gradient Boosting, and LightGBM, intrinsically calculate feature importance as part of their algorithmic process. After the models are trained to predict crop suitability, they provide a score for each feature, indicating the strength of the relationship between that feature and the target variable (the crop type). This scoring helps to

quantify the contribution of each factor, thus guiding decision-making by highlighting which factors should be prioritized for the best agricultural outcomes.

IV. MODELS

A. Multinomial logit and ANOVA Analysis

The F-statistic provides a measure of the variance between the group means relative to the variance within the groups. A high F-statistic suggests that at least one group’s mean is different. The F-statistic, derived from ANOVA (Analysis of Variance) tests, helps in determining whether the differences in crop yields across different treatments (like crop varieties) are statistically significant. For farmers, this is crucial because it indicates whether a particular change in farming practice can lead to different, potentially better, crop outcomes.

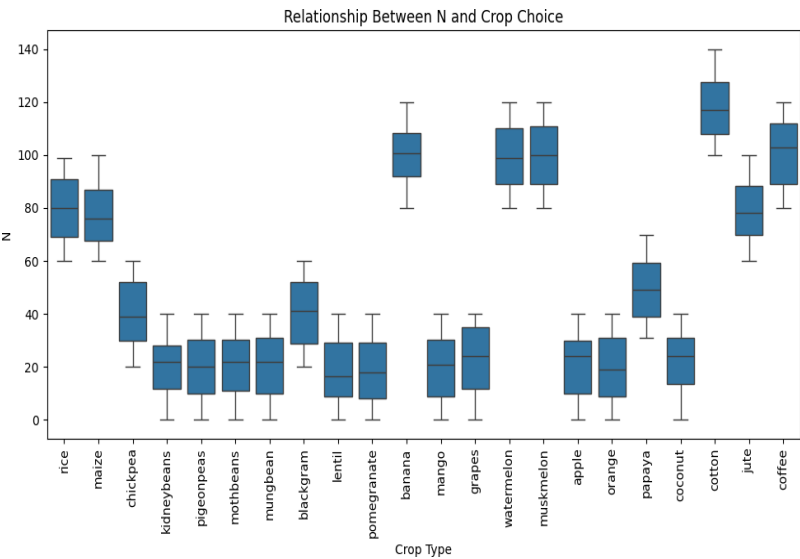


Fig. 1. N and Crop Choice Relation

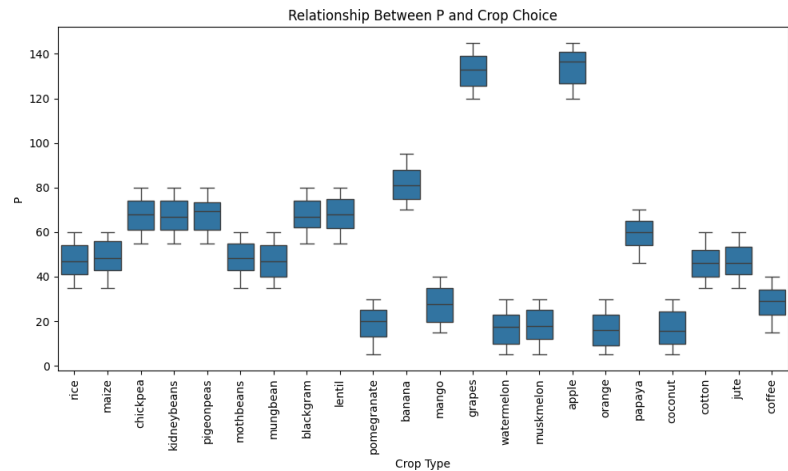


Fig. 2. P and Crop Choice Relation

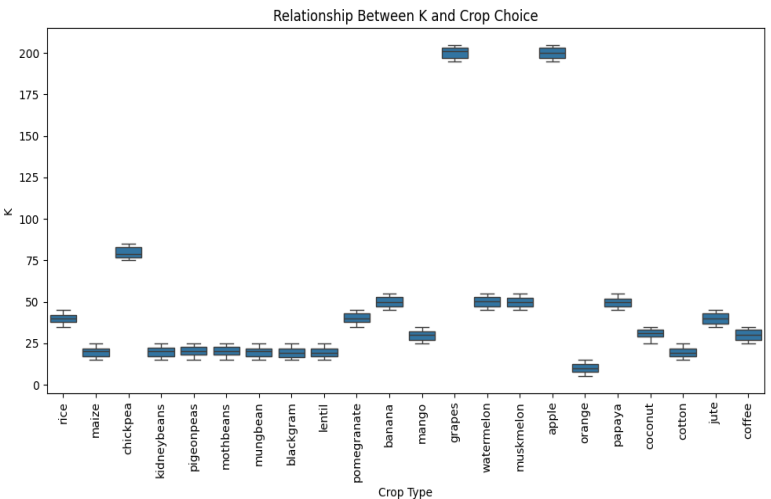


Fig. 3. K and Crop Choice Relation

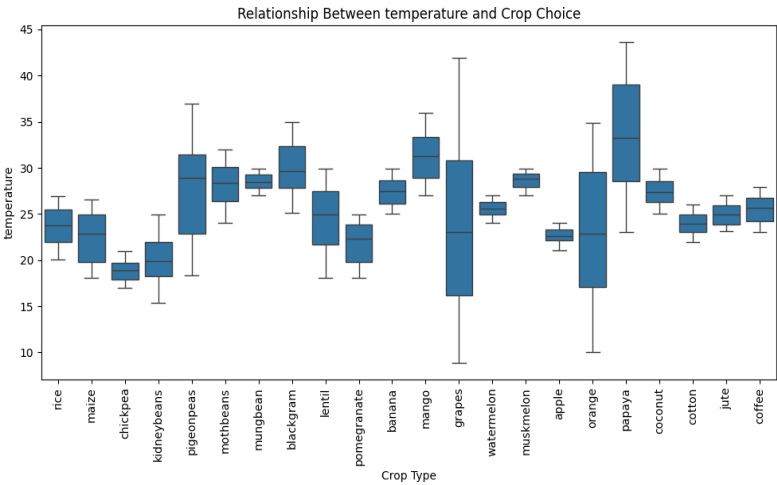


Fig. 4. Temperature and Crop Choice Relation

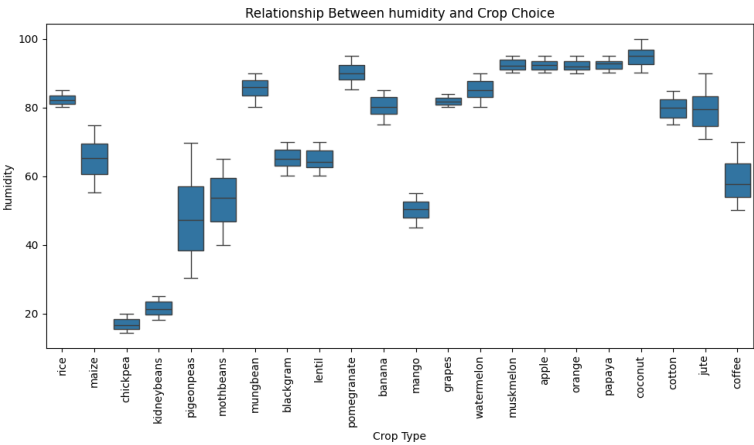


Fig. 5. Humidity and Crop Choice Relation

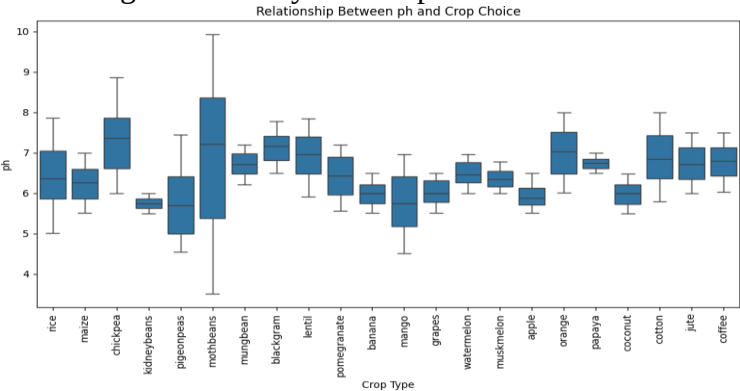


Fig. 6. Ph and Crop Choice Relation

TABLE I  
Multinomial logit- marginal effects on crop choice

| Feature     | F-Statistic | marginal effects<br>(r) | marginal effects<br>(m) |
|-------------|-------------|-------------------------|-------------------------|
| N           | 897.56      | 0.00084***              | 0.00121***              |
| P           | 1885.65     | 0.00031**               | -0.00012                |
| K           | 27238.36    | -0.00020                | -0.00384***             |
| temperature | 102.18      | -0.00504***             | -0.00635***             |
| humidity    | 3103.70     | 0.00029***              | -0.00057**              |
| ph          | 60.34       | -0.01325***             | -0.03058***             |
| rainfall    | 605.52      | 0.00088***              | -0.00049***             |

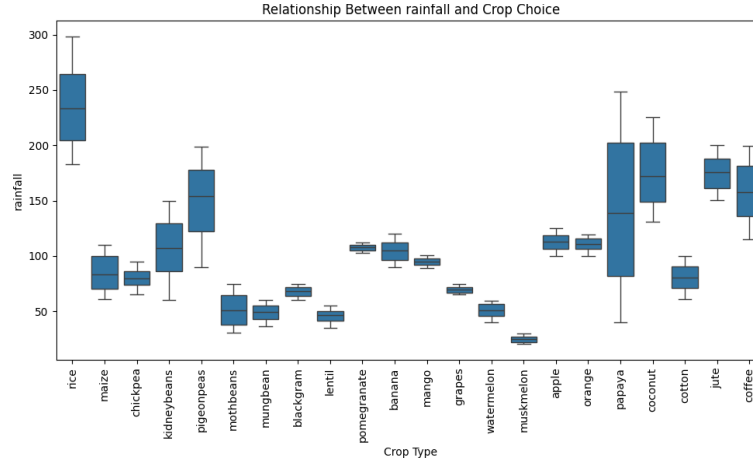


Fig. 7. Rainfall and Crop Choice Relation

- 1) Nitrogen (N): The F-statistic of 897.57 for nitrogen suggests a strong differentiation in nitrogen levels across different crop types, making it a significant factor in deciding suitable crops.
- 2) Phosphorus (P): Similarly, phosphorus, with an F- statistic of 1885.66, is also a key determinant, further emphasizing the role of soil nutrients in crop selection.
- 3) Potassium (K): With the highest F-statistic of 27238.36, potassium levels vary considerably, signifying its crucial influence on crop viability.
- 4) Temperature: A lower F-statistic of 102.19 for temperature, while still significant, may imply less variability across crops compared to soil nutrients but is nonetheless a vital environmental factor.
- 5) Humidity: The F-statistic of 3103.71 indicates that humidity significantly affects crop choice, likely due to its impact on plant transpiration and soil moisture.
- 6) pH: Soil pH, with an F-statistic of 60.34, although the lowest, still shows a significant impact on crop selection, affecting nutrient availability and soil biology.
- 7) Rainfall: The F-statistic of 605.53 for rainfall confirms its substantial effect on crop selection, critical for water supply and influencing irrigation practices.

These results collectively guide agricultural decision- making by identifying which environmental and soil factors should be given priority when selecting crops. Such data- driven insights can lead to more targeted and efficient agricultural practices, optimizing yields and sustainability.

TABLE II  
RANDOM FOREST MODEL PREDICTION ON CROPS

| Crop   | precision | recall | f1-score |
|--------|-----------|--------|----------|
| cotton | 1         | 1      | 1        |
| jute   | 0.920     | 1      | 0.958    |
| lentil | 0.916     | 1      | 0.956    |
| maize  | 1         | 1      | 1        |
| rice   | 1         | 0.894  | 0.944    |

## B. Marginal Effects-Multinomial Logit

We estimate p-values (reported in table I: \*\*\*significant at 1 percent level and \*\* significant at 5 percent level. Here, r refers to crop rice and m refers to crop maize) for each predictor variable and as per the multinomial logit model, most explanatory variables are statistically significant (table I). So, there's evidence that each of these has an independent effect on the probability of a crop choice selection (rather than just a difference observed due to chance). We also calculate the average marginal effects for each explanatory variable to see its effect on crop choice selection. Key results (marginal effect) indicate that nitrogen, rainfall, and humidity have a positive and significant effect on rice production. However, temperature and soil pH have a negative and significant effect on rice production. Phosphorus is positively impacting rice production too, however, at a 5 percent significant level. Potassium does not affect rice production. Additionally, Key results (marginal effect) indicate that nitrogen has a positive and significant effect on maize production. However, potassium, temperature, rainfall, and soil pH have a negative and significant effect on maize production. Humidity is negatively impacting maize production too, however, at a 5 percent significant level. Phosphorous does not affect maize production.

## C. Machine Learning Model

- 1) Random Forest Model: The provided data (in Table II) showcases the efficacy of a Random Forest model in predicting the most suitable crops (e.g., cotton, jute, lentil, maize, rice) based on agro environmental conditions. Model accuracy is higher for cotton, maize and rice compared to jute and lentil. With precision, recall, and F1-scores predominantly at a perfect score of 1 for cotton and maize, the model demonstrates remarkable accuracy and reliability. This high level of performance implies that when the model predicts a certain crop type based on given input features, it is almost always correct (precision), and it successfully identifies nearly all actual instances of each crop type (recall).

Particularly impressive is the model's ability to maintain high scores even for crops like jute, lentil, which present slightly more challenging cases. For rice, while recall is slightly lower at 0.89, indicating some instances were missed, the model still achieves a respectable F1-score of 0.94.

- 2) Gradient Boosting Model: The provided Gradient Boosting model's performance metrics (see table III) across various crops indicate a high level of precision, recall, and F1-score.

TABLE III  
GRADIENT BOOSTING MODEL PREDICTION ON CROPS

| Crop   | precision | recall | f1-score |
|--------|-----------|--------|----------|
| cotton | 0.944     | 1      | 0.971    |
| jute   | 0.821     | 1      | 0.901    |
| lentil | 0.916     | 1      | 0.956    |
| maize  | 1         | 0.952  | 0.975    |
| rice   | 1         | 0.789  | 0.882    |

TABLE IV  
LIGHT GBM MODEL PREDICTION ON CROPS

| Crop   | precision | recall | f1-score |
|--------|-----------|--------|----------|
| cotton | 0.944     | 1      | 0.971    |
| jute   | 0.920     | 1      | 0.958    |
| lentil | 0.846     | 1      | 0.916    |
| maize  | 1         | 0.952  | 0.975    |
| rice   | 1         | 0.894  | 0.944    |

The accuracy is 0.981. While most crops are predicted with high scores, indicating a strong model fit, a few such as cotton, jute, lentil, maize, have slightly lower precision or recall, impacting their F1-scores. The overall high scores imply that the model can effectively utilize environmental data to predict suitable crops, which is invaluable for optimizing agricultural outputs. The ability to accurately match crop types to environmental conditions using such a model can lead to better-informed decisions, potentially increasing yield and improving resource management in agricultural practices.

- 3) Light GBM Model: The Light GBM model's performance metrics shown in table IV reflect a highly effective (efficiency 0.986) prediction capability, with the majority of crops achieving higher precision, recall, and F1-scores. This suggests that the model can predict with higher accuracy whether a crop type should be planted or not, based on given agro environmental conditions. For certain crops like cotton, jute, lentil, there are minor deviations from a perfect score, indicating room for model optimization. However, these scores are still high, suggesting that the model predictions are largely accurate and reliable.

#### D. Feature Importance

The feature importance data across three models—Random Forest, Gradient Boosting, and LightGBM—provides insight into which environmental factors are most predictive of crop type. While the order of importance varies slightly between models, all three identify rainfall and humidity as the top two factors, underscoring their critical role in agricultural modeling. In the Random Forest model, potassium (K) emerges as the third most important feature, which may indicate its effectiveness in capturing complex interactions within the data. The Gradient Boosting model places a slightly greater emphasis on humidity than rainfall, and potassium (K)

remains a significant predictor, though less so than in the Random Forest model. LightGBM gives similar prominence to rainfall and humidity but ranks phosphorus (P) higher than potassium

TABLE V  
FEATURE IMPORTANCE OF DIFFERENT ML MODELS

| Feature     | importance(RF) | importance (GB) | Importance(LGBM) |
|-------------|----------------|-----------------|------------------|
| rainfall    | 0.227          | 0.242           | 0.226            |
| humidity    | 0.211          | 0.232           | 0.225            |
| K           | 0.181          | 0.190           | 0.132            |
| P           | 0.143          | 0.120           | 0.108            |
| N           | 0.108          | 0.119           | 0.107            |
| temperature | 0.075          | 0.055           | 0.105            |
| ph          | 0.052          | 0.038           | 0.093            |

(K), potentially reflecting the model's handling of feature relationships differently due to its gradient-based learning.

Temperature and pH values have lower importance across all models, suggesting they may have less direct influence on crop type prediction compared to other environmental factors. The consistency in feature importance rankings across models reinforces the robustness of the findings and the potential utility of these models in accurately predicting suitable crops based on environmental conditions.

#### E. Boxplot Visualizations

Seven boxplots offer a visual summary of the various environmental preferences and requirements for a range of agricultural crops.

- 1) Rainfall: The Fig. 7 plot reveals that crops like rice and cotton are high rainfall consumers, with their growth optimally supported in regions experiencing heavy precipitation. Rice particularly shows a wide range of tolerance to rainfall intensity.
- 2) Soil pH: The Fig. 6 chart indicates that most crops prefer a soil pH close to neutral, although crops like rice and banana can tolerate a broader pH spectrum. This suggests flexibility in soil acidity or alkalinity for these crops, while others might need more controlled pH conditions.
- 3) Humidity: The Fig. 5 boxplot suggests that rice again has higher humidity requirements, while crops such as chick-peas and pigeon peas can thrive in lower humidity, demonstrating varied adaptability to atmospheric moisture.
- 4) Temperature: The Fig. 4 plot shows that crops like banana and papaya thrive in warmer temperatures, with their optimal growth occurring in warmer climates. Conversely, other crops display a wider range of temperature tolerances, indicating adaptability to different climatic zones.

- 5) Potassium (K): The Fig. 3 boxplot displays a general trend of moderate potassium needs across various crops, with none showing extreme variability or outliers, suggesting that potassium is a less limiting factor in crop choice.
- 6) Phosphorus (P): The Fig. 2 plot reveals a relatively uniform requirement for phosphorus across the crops, with no extreme needs indicated. This nutrient seems to be necessary but not differential so across different types of crops.
- 7) Nitrogen (N): The Fig. 1 boxplot illustrates that nitrogen is a critical nutrient with a broad range of requirements among crops. Rice stands out again, with a particularly high need for nitrogen, which is essential for its growth.

In summary, these boxplots together paint a picture of the multifaceted nature of crop cultivation, where each crop has its specific environmental and nutritional niche. While some crops are versatile and can grow under a range of conditions, others have precise needs that must be met for optimal growth. This data can guide farmers in selecting the right crop for their region based on local environmental conditions and soil nutrient availability.

## CONCLUSION

As per the marginal effects of the multinomial logit model, Nitrogen has a positive and statistically highly significant impact on both crops (rice and maize). Even though rainfall and humidity have a positive effect on rice, their effect is negative on maize. Factors like temperature and soil pH have negative and significant effects on both crops as well. Potassium does not affect rice production whereas phosphorous does not affect maize production. We see rainfall and humidity are the most critical environmental factors to consider when choosing crops, as reflected by their strong feature importance in all machine learning models used. Soil nutrients, particularly potassium (K), also play a significant role, evidenced by their high importance values and the highest F-statistic. Phosphorus (P) and nitrogen (N) follow in importance and are essential but less influential than K. Temperature and pH have the least impact but are still relevant factors. The high-performance metrics (precision, recall, F1 scores) of the Random Forest, Gradient Boosting, and LightGBM models confirm the reliability of these factors in predicting suitable crop choices. Thus, for effective crop selection, a focus on managing water resources and soil nutrition will be essential, while also considering the specific temperature and pH requirements of the crops. In future, an application could involve the development of an AI-driven decision support system for farmers, utilizing real-time environmental and soil data.

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